

Directed Study Syllabus: Advanced studies in Counterfactual Analysis.

Emory University: Department of Economics

09 January 2025

1 About

This Directed study is designed to be an intensive reading in econometrics on a topic not covered in a regular course at Emory University. This course explores a variety of topics in counterfactual analysis with applications in causal inference literature. The course is based on student presentations and discussions among participants (students and faculty invited).

2 Course Description

This reading study session explores the theoretical foundations, methodological advancements, and practical applications of counterfactual analysis in causal inference in depth. The course builds from the basic principles to advanced topics, incorporating sensitivity analyses, handling nonstationary data, and implementing counterfactual models.

3 Content

- Topic 1: **The Journey of Counterfactual Analysis.** This session lays the foundation for the course by introducing causal inference through counterfactual reasoning. It covers the conceptual basis of causality from both the structural and potential outcomes traditions, drawing on the perspectives of Pearl [Pearl \(2009\)](#) and Imbens and Rubin [Imbens and Rubin \(2015\)](#). The session also emphasizes the role of thought experiments in econometric causality [Heckman and Pinto \(2024\)](#), discusses the problem of missing counterfactuals, selection bias, and confounding, and provides a broad overview of major causal inference designs. For a review of Pearl's framework, see also [McDonald \(2002\)](#).
- Topic 2: **From Causal Inference to Dynamic Systems and an Introduction to Synthetic Control.** This session extends counterfactual analysis to time-series settings where treatments and outcomes evolve dynamically. It introduces the potential-outcomes framework for dynamic systems, defines dynamic causal effects and impulse response functions, and discusses the conditions under which these objects admit a causal interpretation [Angrist et al. \(2018\)](#); [Ballinari and Wehrli \(2024\)](#). The session then motivates synthetic control methods for aggregate interventions in settings with limited overlap, and introduces their basic setup, estimation, empirical implementation, and placebo-based inference [Abadie and Gardeazabal \(2003\)](#); [Abadie et al. \(2010\)](#).
- Topic 3: **Counterfactual Imputation under Predictable Errors.** This session examines econometric challenges that arise when standard counterfactual methods leave systematic structure in the residuals. After revisiting

the limitations of traditional synthetic control methods—including sensitivity to poor pre-treatment fit, overfitting in high-dimensional settings, nonstationarity, and dependence across units and time—the session introduces the Artificial Counterfactual (ArCo) estimator as a model-based framework for counterfactual prediction in high-dimensional panel time-series data. The main focus then turns to the problem of predictable imputation errors, as emphasized by [Gonçalves and Ng \(2024\)](#) and the related discussion in [Medeiros \(2024\)](#). In particular, the session studies how serial correlation, cross-sectional dependence, and model misspecification can distort counterfactual prediction and inference, and how Practical Unbiased Predictors (PUP/PLUP) can be used to correct for these nonspherical error structures. The discussion also compares these ideas to related approaches such as FarmTreat and ArCo, and considers extensions based on dynamic systems such as VAR and factor-augmented models.

- Topic 4: **Research Proposal: Extending Synthetic Control to Dynamic Treatment Effects.** This session presents a research proposal motivated by a limitation of traditional synthetic control and related counterfactual-imputation methods: they typically estimate a sequence of contemporaneous treatment effects under a framework of continued treatment exposure, rather than the dynamic causal effects of a one-time intervention. The proposed project develops a panel-data framework for estimating how the effects of a one-time shock propagate over time. Building on the dynamic potential-outcomes literature, the session defines an estimand for dynamic treatment effects, discusses identification through counterfactual imputation, and proposes a two-stage estimation strategy based on predictive modeling with latent factors and autoregressive dynamics. The presentation also outlines sufficient conditions for consistency and asymptotic normality, and evaluates the proposed estimator through simulation exercises under iid and correlated error structures. Empirical motivation is drawn from applications such as natural disasters and macroeconomic stimulus shocks, where treatment occurs once but its effects evolve dynamically over time. [Abadie et al. \(2010\)](#); [Medeiros \(2024\)](#)
- Topic 5: **Difference-in-Differences with Multiple Time Periods.** This session studies modern Difference-in-Differences methods for settings with repeated time periods and variation in treatment timing. It begins with the canonical 2×2 DiD framework, reviewing the potential outcomes setup, the no-anticipation assumption, and the parallel trends assumption as the key ingredients for identifying the average treatment effect on the treated. The discussion then extends to event-study and staggered-adoption designs, emphasizing how treatment effect dynamics and cohort-specific timing complicate identification in empirical applications. A central focus of the session is the limitation of traditional two-way fixed effects estimators, which may recover non-convex weighted averages of treatment effects and can even assign negative weights under heterogeneous treatment effects. To address these issues, the session introduces the Callaway and Sant’Anna framework based on group-time average treatment effects $ATT(g, t)$, along with identification under alternative comparison groups, the role of covariates, aggregation schemes for summarizing treatment effects, and methods for estimation and inference.

4 Proposed Timeline

Meetings are proposed for Sundays from 1:00 pm to 2:00 pm, either in person or online, as arranged by the student and faculty. Deviations from this schedule may be announced in advance. The following gives a rough overview of the meetings.

- 26 January 2025: Introduction and course expectations
- 9 February 2025: Topic 1
- 23 February 2025: Topic 2

- 9 March 2025: Topic 3
- 6 April 2025: Topic 4
- 20 April 2025: Topic 5

References

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